more memory on slower cores. GPUs like the 8600/8500/7600 series generally perform optimally with 256 MB of fast memory. What card manufacturers do is up the amount of video memory to, say, 512 MB, but use DDR2 RAM where DDR3 would normally be used. Such a card will not have much of an advantage over a card with, say, half the memory—simply because the mid-range and entry-level graphics cores don't have sufficient horsepower to keep all that memory busy. And due to slower memory speeds, there's less bandwidth available when it may really be needed. In some cases (when a game demands a larger frame buffer) 512 MB may be marginally better than 256 MB, but for the most part, a fast 256 MB will outperform slower 512 MB of video memory on these lower GPUs.

How They Performed

Despite the fact that the GTS and GT versions of NVIDIA's G84 (8600 series) are only separated by sheer clock speeds, there is quite a difference between the performance of cards based on these chipsets. This bespeaks good clock scaling—a good thing. The 8500GT based cards aren't in the same league as their elder siblings—there's a good bit of difference in the architecture, and it shows. We were surprised to see the ASUS EN8600GT outperform all the other 8600GT cards, and in fact demolish the 8600GTS cards. The reason for this seemingly magical performance turned out to be a factory-overclocked core—from 540 MHz to 625 MHz. The Point of View 8600GT seemed to be facing some issues with many of the benchmarks, and puzzlingly, delivered consistently lower frames than other cards. We ran the benchmarks several times and tried driver reinstalls, even swapping other cards and testing... inexplicable!

ASUS' EN8500GT Silent was another standout. It manages to come close to the other 8600GT cards; we initially thought we'd accidentally plugged in another 8600GT—another case of overclocking, this time by a whopping 150 MHz over stock speeds (600 vs. 450 MHz). XFX's PV-T84J-UDF7 (8600GT) is another good performer—it consistently manages to stay ahead of the other 8600GT based cards by tiny margins.

Choices, Choices

If you're looking for an excellent performer that will cost you significantly less than even lower-performing parts, then look no further than the ASUS EN8600GT (Rs 8,800). It matches

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the 8600GTS cards at just 60 per cent of their price. If you have a budget of around Rs 7,000, consider saving a little more for this excellent card. If you need further convincing, we've awarded it our Best Buy Gold award. The overclocked EN8600GTS TOP from ASUS gets our Best Buy Silver, but to be very honest, we'd suggest you stick with the Gold Winner because of its sterling price.

At a lower budget, XFX's PV-T84J-UDF7 will offer all the thrills, but with a little less wind in your FPS sails at a great price of Rs 7,200. In case you're looking for a good card for an HTPC that will allow you to enjoy HD content in all its visual splendor, take a look at Sparkle's 8500GT 256MB—at Rs 4,110, this is component for an HTPC with the occasional casual game thrown in.

ATI's HD2600Pro chipsets don't create too much of a splash—for one, they're slightly overpriced for the performance they deliver. As things currently stand, we can't really recommend any cards based on these chipsets as solutions.

ASUS-EN8600GTS silent
Fast. Silent. Expensive.

Something For Everybody

Contrary to the many industry experts who believed that the *bigger, faster, better* race between NVIDIA and ATI would slow down, this year has actually seen an acceleration of sorts. DX 10 being the wet squib it turned out to be notwithstanding, both companies have announced successors to their current line-up of cards. Rumours abound regarding release dates, but we have sufficient reason to believe NVIDIA has a GeForce 9 series not very far from the retail markets. ATI, too, has announced an HD3000 family, though details are sketchy at best. These cards should have support for DX 10.1—although once again, how they perform on an API that nobody's seen is anyone's guess.

But all this is irrelevant to most of you. The fact is, all this hype and competitiveness has brought around better technology at much cheaper prices. We reckon graphics cards aren't the novelty they were once considered. If you've been holding on to those purse strings for all you're worth, now may just be the time to indulge yourself... !⊠

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Contact Sheet — Graphics Cards

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ASUS	Asus Technology Pvt Ltd	022-6766 8800	http://in.asus.com
BIG	BIG Byte Corporation	022-23894800 /1/2/3/4	www.bigbyteindia.com
Foxconn	CSD EU B.V. (Foxconn-MEIA)	00-971-4-2045266/5276	www.foxconnchannell.com
Galaxy	Technology and Gadgets	91-22-23823331 /2	www.technologyandgadgets.com
GeCube	Tirupati Enterprises	9339207519	www.tirupati.net
Gigabyte	Gigabyte Technology India Ltd	022-30616666	www.gigabyte.com.tw
Leadtek	Zebronics / TopNotch Infotronix	044-26616201	www.zebronics.com
MSI	Micro-Star International	011-41758808	www.msi-india.com
Point of View	Tirupati Enterprises	9339207519	www.tirupati.net
Sparkle	Abacus Peripherals Pvt Ltd	022-40914600	www.abacusperipherals.com
XFX	Rashi Peripherals Pvt Ltd	022-67090909	www.rptechindia.com
Zebronics	Zebronics / TopNotch Infotronix	044-26616201	www.zebronics.com